

Song Search System.

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Project Overview

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Milestone 2

Overview

- Datasets gathered from:
 - Kaggle
 - Last.FM
- Collection
- Preparation
- Characterization

Data Indexing

The documents to be indexed are the result of the data pipeline previously referred, which was collected, prepared, and cleaned for this phase.

Document Definition

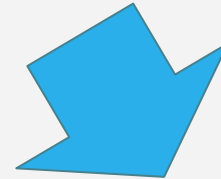
Each song comprises a title, an artist, features, views, and atag, and it is also provided from Last.FM, a play count, number of different listeners, and content of both song and artist. We have some long textual fields, such as the content of the artist and song and also lyrics, which will be useful for these further steps.

Data Indexing

Two Different Schemas



Simple -> No Tokenizers and Filters



Advanced -> Uses a tokenizer and filters

Data Indexing

For Advanced Schema:

- **StandardTokenizerFactory** - Breaks text into words or expressions, to allow a simple rule applying by filters.
- **ASCIIFoldingFilterFactory** - Converts any non-ASCII characters to their ASCII equivalents.
- **LowerCaseFilterFactory** - Converts all uppercase letters to their lowercase version.
- **PortugueseStemFilterFactory** - Reduces Portuguese words to their root form. (Example: casinha -> cas).
- **SynonymGraphFilterFactory** - Expands tokens to include their synonyms. We also defined those synonyms in synonyms.txt
- **FlattenGraphFilterFactory** - Simplifies complex token graphs into a linear sequence for easier processing.

Data Indexing

For Advanced Schema:

Field	Type	Indexed?
lyrics	boostedText	yes
title	boostedText	yes
views	Int	no
artist	shortText	yes
tag	string	yes
features	shortText	yes
lastfm_different_listeners	Int	no
lastfm_playcount	Int	no
lastfm_tags	string	yes
lastfm_content_song	boostedText	yes
lastfm_content_artist	boostedText	yes
lastfm_image_artist	string	no
lastfm_image_album	string	no
album	string	yes

Data Retrieval

Parameter	Value
q	religião
q.op	AND
fl	*, score

Table 2: Common query parameters

Parameter	Value
qf	lyrics^ 4 title^ 2 lastfm_content_artist^ 3 tag^ 10 artist
pf	lyrics^ 6 lastfm_content_artist^ 6 title^ 10
ps	3

Table 3: Boosted schema parameters

Data Evaluation

Used Metrics

Accuracy: Indicates the percentage of things that were successfully and unsuccessfully retrieved from the total number of items.

Recall: The percentage of pertinent papers that were recovered from the whole collection of pertinent documents.

Precision: The percentage of pertinent documents returned.

Precision at 10 (P@10): A targeted precision statistic that assesses how relevant the top 10 documents that were retrieved are.

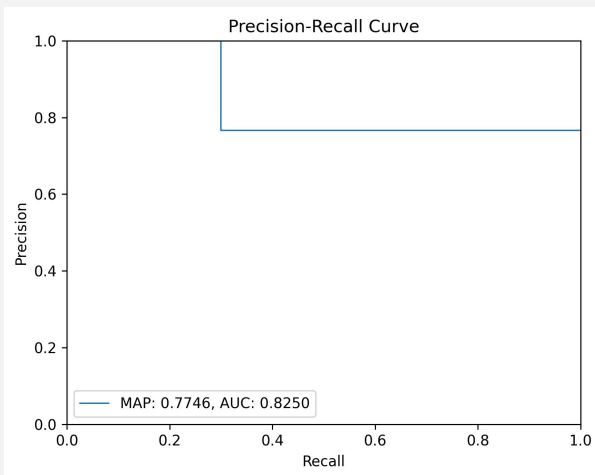
Average Precision (AvP): Highlights the ranking order of pertinent things by calculating the average precision values at each time a relevant document is obtained.

Mean Average Precision (MAP): The mean accuracy of all the evaluation's queries.

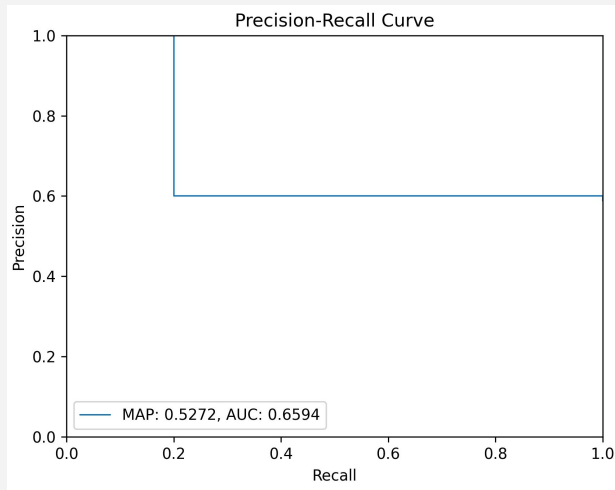
Query

Songs with lyrics related to religion.

Query 1 - Boosted Schema



Query 1 - Simple Schema



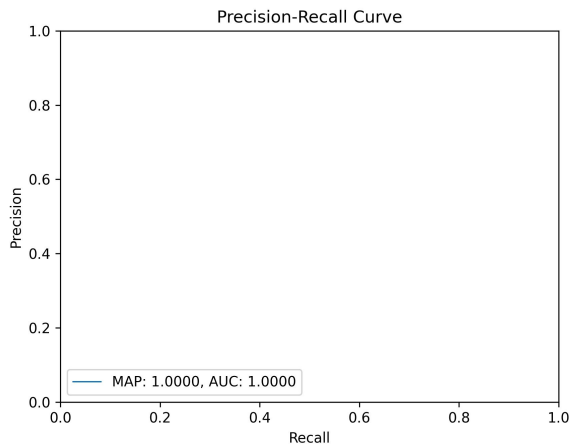
Query 1 - Results

Rank	Simple System	Advanced System
P@10	0.5000	0.6000
P@30	0.3333	0.7333
AvP	0.5272	0.7317

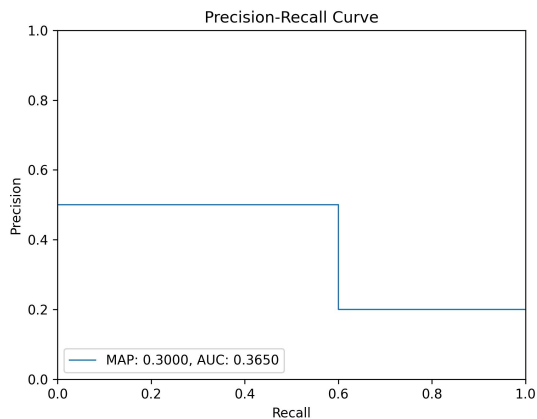
Query

Songs whose artist has hit number 1 in the charts and is Portuguese.

Query 2 - Boosted Schema



Query 2 - Simple Schema



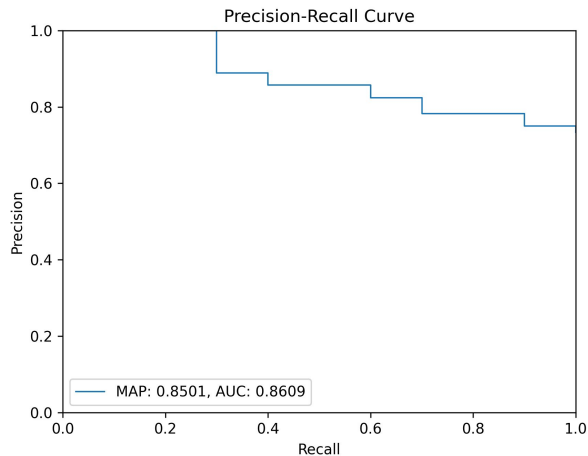
Query 2 - Results

Rank	Simple System	Advanced System
P@10	0.3000	1.0000
P@30	0.1333	0.6667
AvP	0.3000	1.0000

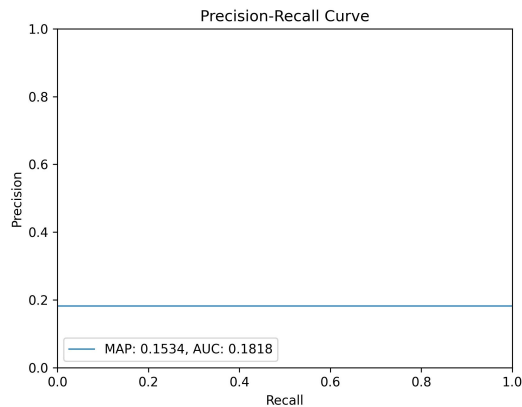
Query

Songs whose artist is female that talk about empowerment.

Query 3 - Boosted Schema



Query 3 - Simple Schema



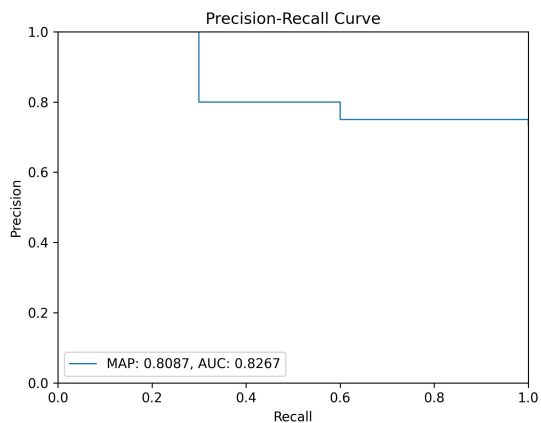
Query 3 - Results

Rank	Simple System	Advanced System
P@10	0.1000	0.8000
P@30	0.0667	0.7333
AvP	0.1534	0.8501

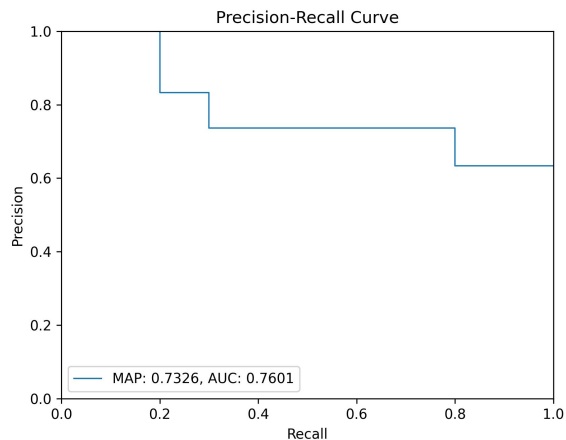
Query

Songs that talk about reflection or love.

Query 4 - Boosted Schema



Query 4 - Simple Schema



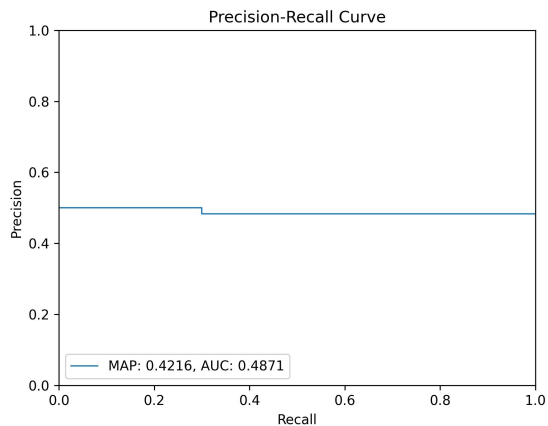
Query 4 - Results

Rank	Simple System	Advanced System
P@10	0.6000	0.8000
P@30	0.6333	0.7333
AvP	0.7326	0.8087

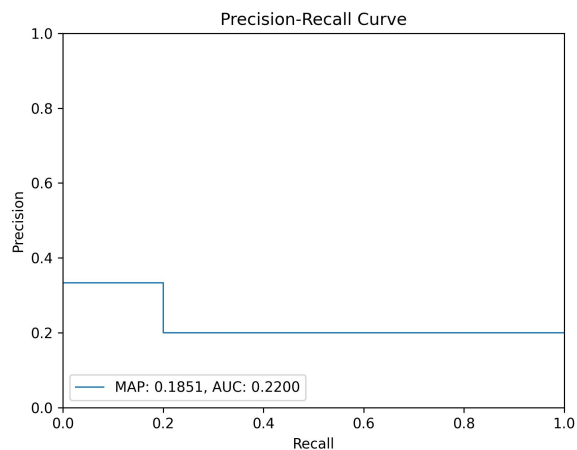
Query

Pop songs that talk about the past.

Query 5 - Boosted Schema



Query 5- Simple Schema



Query 5 - Results

Rank	Simple System	Advanced System
P@10	0.1000	0.4000
P@30	0.2000	0.4667
AvP	0.1851	0.4216

Queries

Final Conclusions

Global	Simple System	Advanced System
MAP	0.3848	0.7632

A Year To Listen: Song Search System.

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